

SURYANSH SHRIVASTAVA

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SUMMARY

Research-oriented engineer building LLM data, evaluation, and agent infrastructure. Production experience with controlled synthetic-data generation, LLM-as-a-judge evaluation systems, rubric design and validation, Docker-based agent evaluation environments, and reinforcement learning. Comfortable owning ambiguous, zero-to-one problems end to end.

EXPERIENCE

Microsoft — Search, Assistant & Intelligence

July 2024 – Present

Applied Scientist

Hyderabad, India

- Own M365 Copilot's SLM prompt-autocompletion experience end to end: built the initial proof-of-concept by integrating SLM outputs into the existing search and indexing architecture, then generated 50K+ high-quality synthetic prompts via controlled data generation, explicitly shaping intent and prefix distributions to match production traffic; fine-tuned a 0.6B-parameter SLM (Qwen3) for real-time completion and deployed to production serving 3M+ users, improving Click Conversion Rate by 150%.
- Designed an LLM-as-a-judge evaluation framework for generation quality and Responsible AI checks; built an auto-research loop that iteratively proposes, tests, and refines judge rubrics against human-labeled data, improving agreement with human annotators to Cohen's $\kappa = 0.68$.
- Led ranking and relevance experimentation for historical prompt suggestions via A/B testing, contributing to a 65% CCR improvement across M365 Copilot autocompletion experiences.
- Built LLM-powered internal agents automating partner onboarding and prompt-authoring workflows, reducing manual effort and accelerating iteration speed.
- Parallelized and optimized constraint-index lookups to cut production service latency by 37%; built functional test suites and scenario-based monitoring dashboards for reliability and observability across production workflows.

Adobe — Media & Data Science Research Lab

May 2023 – Aug 2023

Research Scientist Intern

Noida, India

- Benchmarked and fine-tuned multimodal foundation models (LLaVA, MiniGPT-4, BLIP-2, LayoutLM) for document layout understanding using LoRA/QLoRA and distributed multi-GPU training (DP/DDP) on Linux clusters.
- Designed layout-aware pretraining tasks and few-shot inference experiments to inject layout knowledge into language models, improving average IoU on layout parsing from 22% to 48% over baseline.

IIT Delhi — Networked Autonomous Systems Lab

May 2022 – Jul 2022

Research Intern

New Delhi, India

- Built an autonomous-driving RL environment on the SUMO framework with occupancy grids and the Intelligent Driver Model for human-driven traffic; defined observation, state, action, and reward spaces and trained a PPO agent over a bifurcated observation space of local and network-assisted views.

PROJECTS

Smol Test Harness — Evaluation Framework for Coding Agents | Python, Docker, Linux, pytest

GitHub · July 2026 – Present

- Built a Python + Docker harness for running and comparing 5 coding-agent CLIs (Claude Code, OpenCode, Copilot CLI, Codex, Pi) on repeatable tasks inside isolated Linux containers.
- Designed an arrange-act-score workflow with 7 reusable evaluation patterns covering bug fixing, feature development, test writing, schema mapping, scorer design, and repository question answering; used hidden tests and phase-separated scoring to prevent answer leakage and improve evaluation reliability.
- Built reliability tooling to distinguish model failures from infrastructure failures: model health checks, phase timeouts, temporary credential mounts, and concurrent execution; captured score, token usage, runtime, and execution status across repeated runs with comparable JSON/CSV exports.
- Wrote unit and Docker integration tests covering orchestration, metric aggregation, failure handling, and container cleanup.

Mo(ve)ments — AI Presentation-Preparation Platform | Next.js, TypeScript, tRPC, PostgreSQL, Claude API GitHub · Live · 2026

- Built and deployed a full-stack platform that transforms a topic and uploaded reference documents into structured “moments” — atomic presentation units pairing slide content with a speaker script, timing, and narrative beat — editable inline and exported to PPTX with speaker notes.
- Engineered the end-to-end LLM generation pipeline: document ingestion (magic-byte validation, PDF/DOCX text extraction, overlapping-window chunking), prompt assembly with cached system prompts, structured JSON generation, and schema validation, with retry/backoff and per-request token cost logging.
- Implemented an automated source-verification QC layer that fuzzy-matches every generated claim against ground-truth document text, flagging hallucinated or unsupported citations before content reaches the user.
- Designed a conversational co-director agent that revises slides and scripts through an XML-based edit protocol — parsed, validated, and applied to persistent state — with prompt-injection defenses treating all uploaded content as untrusted data.
- Shipped production infrastructure: Supabase Postgres with row-level security for per-user data isolation, email + Google OAuth, debounced autosave, rate limiting, and Sentry error monitoring on Vercel.

Airy8 — Discovery Platform for AI Tools | TypeScript, Node.js, PostgreSQL, Azure airy8.com · July 2025 – Present

- Solo-built and shipped a production AI-tool discovery platform with search, ranking, and analytics; developed REST APIs on Node.js/Express over a PostgreSQL + Prisma data layer with schema migrations and indexing.
- Implemented OAuth 2.0 + JWT authentication with role-based access control, rate limiting, and API protection; deployed on Azure App Service with CI/CD via GitHub Actions.

Vasuki — Reinforcement Learning Environment | Python, OpenAI Gym, OpenCV GitHub · Dec 2021 – Jan 2022

- Designed a two-agent grid RL environment from scratch and registered it with OpenAI Gym: modeled state and action spaces, engineered the reward distribution, and implemented a DQN agent based on the Atari DQN architecture, with an OpenCV interface for visualizing agent trajectories.

EDUCATION

Indian Institute of Technology, Guwahati 2020 – 2024
B.Tech, Electronics & Electrical Engineering Guwahati, India

- Relevant coursework: Reinforcement Learning, Data Structures & Algorithms, Linear Algebra, Probability & Random Processes, Game Theory; Deep Learning and NLP Specializations (DeepLearning.AI).

ACHIEVEMENTS

- **Codeforces** (three_nil): contest rating 1906; peak 2040 (**Candidate Master**).
- **JEE Advanced**: All India Rank 1760 among 150,000+ candidates (2020); **JEE Mains**: Rank 2796 among 1M+ candidates.
- **KVPY Scholar**, awarded by the Government of India and Indian Academy of Sciences for research aptitude (2019).
- **NSEA**: national top 1% in the Astronomy Olympiad (2018); **RMO**: qualified for INMO, top 30 in state Mathematics Olympiad (2017).
- **IITG.ai Hackathon**: 1st place for a deep-learning solution (2021).

TECHNICAL SKILLS

LANGUAGES	Python, C++ , TypeScript, SQL
AI & EVALUATION	LLM fine-tuning (LoRA/QLoRA, DDP), synthetic-data generation, LLM-as-a-judge, benchmark design, rubric development, evals, reinforcement learning (DQN, PPO)
FRAMEWORKS	PyTorch, Hugging Face, OpenAI Gym, scikit-learn, OpenCV, TensorFlow
INFRASTRUCTURE	Docker, Linux, pytest, Git, GitHub Actions, CI/CD, Azure, PostgreSQL, REST APIs
EXPERIMENTATION	A/B testing, metric design, validation workflows, Pandas, NumPy